



RECOVERY & REPAIR

LL-37

LL-37 – 5 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	Cathelicidin antimicrobial peptide LL-37
SYNONYMS / ALIASES	LL-37; hCAP-18 C-terminal fragment; CAMP (gene product, mature C-terminal peptide)
CAS NUMBER	154947-66-7
MOLECULAR FORMULA	C205H340N60O53
MOLECULAR WEIGHT	Approximately 4493.3 g/mol
AMINO ACID SEQUENCE	LLGDFFRKSKKEKIGKEFKRIVQRIKDFLRNLVPRTES (37 residues)
SOURCE	Synthetic (solid-phase peptide synthesis); sequence corresponds to the mature antimicrobial fragment of human cathelicidin (CAMP)

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=99% by HPLC (confirm against batch COA)
CONTAINER	5 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, stable approximately 14 days in bacteriostatic water
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only); LL-37 is amphipathic and may adsorb to plastic surfaces - use polypropylene or siliconized labware
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	Highly cationic and amphipathic; can self-aggregate and adsorb to surfaces. Avoid repeated freeze-thaw cycles.

STRUCTURAL NOTES

LL-37 is a 37-amino-acid synthetic peptide corresponding to the mature antimicrobial fragment of human cathelicidin (gene CAMP, precursor hCAP-18). It is classified as a member of the cathelicidin family of antimicrobial peptides (AMPs). Structurally, it adopts an amphipathic alpha-helical conformation in membrane-mimetic environments, with a net positive charge of +6 at physiological pH driven by its lysine and arginine residues. The N-terminal "LL" dipeptide gives the peptide its common name.

SELECTED PEER-REVIEWED REFERENCES

1. Durr U.H. et al. - *LL-37, the only human member of the cathelicidin family of antimicrobial peptides* - Biochimica et Biophysica Acta, 2006 - DOI: 10.1016/j.bbame.2006.03.030
2. Vandamme D. et al. - *A comprehensive summary of LL-37, the factotum human cathelicidin peptide* - Cellular Immunology, 2012 - DOI: 10.1016/j.cellimm.2012.08.012
3. Kahlenberg J.M., Kaplan M.J. - *Little peptide, big effects: the role of LL-37 in inflammation and autoimmune disease* - Journal of Immunology, 2013 - DOI: 10.4049/jimmunol.1302005

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

COMPLIANCE DISCLAIMER

This product is sold by Aligned Aminos for laboratory research use only (RUO). It is not a drug, dietary supplement, food, cosmetic, or medical device. It has not been approved by the FDA for any therapeutic, diagnostic, or preventative use. It is not intended for use in humans or animals, and may not be administered, ingested, applied, or otherwise introduced into any living organism. Purchasers represent that they are qualified researchers or laboratory professionals and assume full responsibility for compliance with all applicable laws and institutional review requirements governing the handling of research chemicals.

Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.